Finite-p Hausdorff Spaceability for Prescribed Graph Dimension

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Problem

In Liu–Shi–Zhang, *On strong spaceability of continuous functions and fractal dimensions*, the following question appears at

`downloads/extracted/2605.25037v2/Onspaceability.tex:1358`.

For $1 < s \leq 2$, put
$$H_s[0,1] = \{f \in C[0,1] : \dim_H G_f([0,1]) = s\}.$$
The question is whether $H_s[0,1]$ is $(p, \mathfrak{c})$-spaceable for every finite integer $3 \leq p < \aleph_0$.

Definition. *Let X be a topological vector space, $A \subset X$, and $p \in \mathbb{N}$. We say that A is $(p, \mathfrak{c})$-spaceable if, whenever E is a p-dimensional linear subspace of X such that*

$$E \subset A \cup \{0\},$$

there is a closed linear subspace $Y \subset X$ with

$$E \subset Y \subset A \cup \{0\}$$

and $\dim Y = \mathfrak{c}$.

Imported result from Liu–Shi–Zhang

We use only the following special case of Liu–Shi–Zhang Proposition 3.4, taking $K = [0,1]$ and $m = 1$.

Proposition (Liu–Shi–Zhang, Proposition 3.4 with $m = 1$)**.** *Let $1 < s \leq 2$, let $p \in \mathbb{N}$, and let $f_1, \ldots, f_p \in H_s[0,1]$ be linearly independent. Assume*

$$\operatorname{span}\{f_1, \ldots, f_p\} \subset H_s[0,1] \cup \{0\}.$$

Then there is a compact set $K_0 \subset [0,1]$ such that

$$\dim_H K_0 = s - 1$$

and, for every nonzero $(a_1, \ldots, a_p) \in \mathbb{R}^p$,

$$\sum_{i=1}^{p} a_i f_i \bigg|_{[0,1] \setminus K_0} \in H_s([0,1] \setminus K_0).$$

Auxiliary facts

We shall use the following standard facts without further comment.

(F1) Hausdorff dimension is countably stable:

$$\dim_H \left(\bigcup_{n=1}^{\infty} A_n\right) = \sup_n \dim_H A_n.$$

(F2) If $K \subset \mathbb{R}$ and $f \in C(K)$, then

$$\dim_H G_f(K) \leq \dim_H K + 1.$$

Indeed $G_f(K) \subset K \times f(K)$, and $f(K) \subset \mathbb{R}$.

(F3) If $K \subset [0,1]$ is compact and $d = \dim_H K$, then K has a full-dimensional point: there is $x_0 \in K$ such that

$$\dim_H(K \cap (x_0 - r, x_0 + r)) = d, \qquad r > 0.$$

Otherwise finitely many neighborhoods of dimension $< d$ would cover K, contradicting countable stability.

(F4) If $E \subset \mathbb{R}$ is an uncountable compact set, then there exists $u \in C(E)$ such that

$$\dim_H G_u(E) = \dim_H E + 1.$$

This is the prevalent maximal-graph-dimension theorem cited as Lemma 2.4 in Liu–Shi–Zhang.

The compact c_0-subspace lemma

Lemma. *Let $K \subset [0,1]$ be compact and let*

$$d = \dim_H K > 0.$$

Then there is a closed subspace $X_K \subset C[0,1]$, isomorphic to c_0, with the following properties.

(i) For every nonzero $h \in X_K$,

$$\dim_H G_h(K) = d + 1.$$

(ii) Every $h \in X_K$ is affine on each connected component of $[0,1] \setminus K$.

Proof. Choose a full-dimensional point $x_0 \in K$. We first construct disjoint pieces of K whose dimensions tend to d.

Let (r_m) be a strictly decreasing sequence of positive numbers tending to 0, chosen so fast that the annuli

$$A_m = K \cap \big((x_0 - r_m, x_0 + r_m) \setminus [x_0 - r_{m+1}, x_0 + r_{m+1}]\big)$$

are pairwise disjoint. Since

$$K \cap (x_0 - r_N, x_0 + r_N) \subset \{x_0\} \cup \bigcup_{m \geq N} A_m,$$

and the left-hand side has Hausdorff dimension d, countable stability gives

$$\sup_{m \geq N} \dim_H A_m = d$$

for every N.

Therefore, after passing to a subsequence of annuli and then using inner regularity of Hausdorff dimension, we may choose compact sets $E_n \subset K$ and open intervals I_n such that

$$E_n \subset I_n, \qquad \overline{I_n} \cap \overline{I_m} = \emptyset \ (n \neq m), \qquad I_n \to x_0,$$

and

$$d_n := \dim_H E_n > d - \frac{1}{n}.$$

Discard finitely many terms if necessary, so that $d_n > 0$ for every n. Then each E_n is uncountable, and $d_n \to d$.

By (F4), for each n there is $\varphi_n \in C(E_n)$ such that

$$\dim_H G_{\varphi_n}(E_n) = d_n + 1.$$

Multiplying by a nonzero scalar if needed, assume $\|\varphi_n\|_\infty = 1$.

Since E_n and $K \setminus I_n$ are disjoint closed subsets of K, Tietze's extension theorem gives a function $\psi_n \in C(K)$ such that

$$\psi_n|_{E_n} = \varphi_n, \qquad \psi_n|_{K \setminus I_n} = 0, \qquad \|\psi_n\|_\infty = 1.$$

The supports of the ψ_n's are pairwise disjoint subsets of K, because the intervals I_n are pairwise disjoint.

Partition $\mathbb{N}$ into pairwise disjoint infinite sets

$$\mathbb{N} = B_1 \cup B_2 \cup \cdots.$$

For each j, choose coefficients $c_{j,n} \in (0, 1]$, $n \in B_j$, such that

$$\sup_{n \in B_j} c_{j,n} = 1 \quad \text{and} \quad c_{j,n} \to 0 \quad (n \to \infty,\ n \in B_j).$$

For example, take one coefficient equal to 1, and make all later coefficients tend to 0.

Define $v_j \in C(K)$ by

$$v_j(x) = \begin{cases} c_{j,n}\psi_n(x), & x \in K \cap I_n \text{ for some } n \in B_j, \\ 0, & \text{otherwise.} \end{cases}$$

The only possible accumulation point of the intervals I_n is x_0, and the coefficients $c_{j,n}$ tend to 0 along B_j. Hence each v_j is continuous at x_0, and it is plainly continuous elsewhere. Also $\|v_j\|_\infty = 1$.

Now define

$$T : c_0 \longrightarrow C(K), \qquad T((a_j)_{j \geq 1}) = \sum_{j=1}^{\infty} a_j v_j.$$

This is a pointwise locally finite sum away from x_0, and continuity at x_0 follows from $a_j \to 0$ together with $v_j(x_0) = 0$. Since the supports of the v_j's are pairwise disjoint,

$$\|T(a)\|_\infty = \sup_j |a_j|.$$

Thus T is an isometric embedding and $T(c_0)$ is a closed subspace of $C(K)$.

If $a = (a_j) \neq 0$, choose j_0 with $a_{j_0} \neq 0$. For every $n \in B_{j_0}$, on E_n we have

$$T(a) = a_{j_0} c_{j_0,n} \varphi_n.$$

Nonzero scalar multiplication does not change Hausdorff dimension, so

$$\dim_H G_{T(a)}(E_n) = d_n + 1, \qquad n \in B_{j_0}.$$

Therefore

$$\dim_H G_{T(a)}(K) \geq \sup_{n \in B_{j_0}} (d_n + 1) = d + 1.$$

The reverse inequality is (F2), so

$$\dim_H G_{T(a)}(K) = d + 1.$$

Finally, let $\mathcal{E}_K : C(K) \to C[0,1]$ be the canonical linear extension obtained by affine interpolation on every connected component of $[0,1] \setminus K$. This operator is an isometric embedding. Put

$$X_K = \mathcal{E}_K(T(c_0)).$$

Then X_K is closed in $C[0,1]$, isomorphic to c_0, satisfies the same graph-dimension conclusion on K, and is affine on every component of $[0,1] \setminus K$. □

Main theorem

Theorem. *Let $1 < s \leq 2$. Then $H_s[0,1]$ is $(p, \mathfrak{c})$-spaceable for every finite $p \in \mathbb{N}$.*

Proof. Fix $p \in \mathbb{N}$. Let $E \subset C[0,1]$ be a p-dimensional subspace such that

$$E \subset H_s[0,1] \cup \{0\}.$$

Choose a basis $f_1, \ldots, f_p$ of E. Applying the imported Liu–Shi–Zhang proposition gives a compact set $K_0 \subset [0,1]$ such that

$$\dim_H K_0 = s - 1$$

and

$$w|_{[0,1] \setminus K_0} \in H_s([0,1] \setminus K_0), \qquad 0 \neq w \in E.$$

Set $d = s - 1$. Since $s > 1$, we have $d > 0$. Apply Lemma to K_0, and let $X_0 = X_{K_0}$. Thus X_0 is closed, $\dim X_0 = \mathfrak{c}$, every nonzero $h \in X_0$ has

$$\dim_H G_h(K_0) = s,$$

and every $h \in X_0$ is affine on the connected components of $[0,1] \setminus K_0$.

We first prove that

$$E \cap X_0 = \{0\}.$$

Suppose $0 \neq w \in E \cap X_0$. Since $w \in E$, the choice of K_0 gives

$$\dim_H G_w([0,1] \setminus K_0) = s.$$

Since $w \in X_0$, the function w is affine on each connected component of $[0,1] \setminus K_0$. The set $[0,1] \setminus K_0$ is a countable union of open intervals, and the graph of an affine function on each interval has Hausdorff dimension 1. Hence

$$\dim_H G_w([0,1] \setminus K_0) \leq 1,$$

contradicting $s > 1$. Thus the intersection is trivial.

Let

$$Y = E \oplus X_0.$$

Because E is finite-dimensional and X_0 is closed, the direct sum Y is a closed subspace of $C[0,1]$. Moreover $E \subset Y$ and $\dim Y = \mathfrak{c}$.

It remains to show that

$$Y \subset H_s[0,1] \cup \{0\}.$$

Let $0 \neq y \in Y$. Write uniquely

$$y = w + h, \qquad w \in E, \; h \in X_0.$$

Assume first that $w \neq 0$. Let

$$U = [0,1] \setminus K_0.$$

On every connected component of U, the function h is affine. Therefore, on each such component J, the map

$$(x,t) \mapsto (x, t + h(x))$$

is bi-Lipschitz on $J \times \mathbb{R}$. Hence

$$\dim_H G_{w+h}(J) = \dim_H G_w(J).$$

Using countable stability over the components J of U, we get

$$\dim_H G_y(U) = \dim_H G_w(U) = s.$$

On the compact set K_0, the general upper estimate gives

$$\dim_H G_y(K_0) \leq \dim_H K_0 + 1 = s.$$

Thus

$$\dim_H G_y([0,1]) = \max\{\dim_H G_y(K_0), \dim_H G_y(U)\} = s.$$

Assume now that $w = 0$. Then $y = h \neq 0$, and Lemma gives

$$\dim_H G_y(K_0) = s.$$

The same upper estimate gives $\dim_H G_y([0,1]) \leq s$, while the previous display gives the reverse inequality. Hence again

$$\dim_H G_y([0,1]) = s.$$

So every nonzero element of Y belongs to $H_s[0,1]$. Since the initial p-dimensional subspace E was arbitrary, $H_s[0,1]$ is $(p, \mathfrak{c})$-spaceable. □

Consequence

Liu–Shi–Zhang also prove that $H_s[0,1]$ is not (α, β)-spaceable for any $\alpha \geq \aleph_0$, regardless of β. Therefore the theorem above gives the positive side for every possible finite initial dimension p, and the negative side for every infinite initial dimension was already known in their paper.

Human-check checklist

- Verify the extraction of the $m = 1$ case of Liu–Shi–Zhang Proposition 3.4.
- Verify the annulus selection in Lemma ; it is the only place where dimension concentration at one point matters.
- Verify the use of affine interpolation $\mathcal{E}_K$, especially that it is isometric and affine on complementary intervals.
- Verify that the countable component argument on $[0, 1] \setminus K_0$ replaces the "nearly locally Lipschitz" lemma without requiring compactness of the domain.